Venkata Lakshmi Parimala Pasupuleti

+1-703-261-5410 | venkatalakshmiparimala@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti |

Fairfax, Virginia (Remote)

PROFESSIONAL SUMMARY

Business analyst and data professional with 9+ years of experience gathering requirements, analyzing workflows, and delivering data-driven recommendations across enterprise SaaS and regulated domains. Built AgentProbe, an open-source chatbot and LLM evaluation toolkit that scores AI outputs on factual accuracy, hallucination rate, retrieval quality, completeness, and compliance - directly applicable to structured chatbot response evaluation. Proficient in SQL, Python, Power BI, Tableau, and Excel; experienced with Agile/Scrum across 120+ delivery sprints. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

CORE COMPETENCIES

Business Requirements Analysis | Chatbot & LLM Output Evaluation | Process Mapping & Gap Analysis | SQL & Data Analysis | Power BI / Tableau / Excel | Stakeholder Communication | Functional Specifications & Documentation | Agile / Scrum | Cross-functional Team Coordination | KPI Tracking & Reporting | User Acceptance Testing | Python (Pandas, NumPy)

EXPERIENCE

Intuit

Jun 2026 - Present

Software Engineer · AI Applications · Virginia, USA

- Partner with Finance and product teams to **gather and translate ambiguous business requirements** into functional AI application specs - defining evaluation criteria, acceptance conditions, and success KPIs before build.
- Build **structured evaluation frameworks** (LLM-as-judge + golden datasets) to score AI output quality across dimensions including accuracy, completeness, and compliance - track regression metrics across releases.
- Create dashboards and reports to communicate AI system performance and workflow improvement metrics to technical and non-technical stakeholders.
- Coordinate with cross-functional teams (product, compliance, engineering) to track project progress, surface risks, and ensure delivery against defined criteria.

George Mason University - College of Science

Aug 2024 - Aug 2025

Research Associate · Data Analysis & Reporting · Fairfax, Virginia

- Analyzed multi-source datasets (190+ countries, WHO + UK Road Safety) - documented data requirements, identified gaps, built preprocessing pipelines and regression models ($R^2 = 0.977$).
- Built **Tableau, ggplot2, and Quarto dashboards** translating complex findings into actionable faculty recommendations - boosted stakeholder comprehension by 40%.
- Designed anomaly detection process (Isolation Forest) that replaced manual QA reviews across 3 research projects - cut review time by 70% while improving accuracy.
- Maintained detailed process documentation, query logs, and methodology notes for reproducibility and handoff.

Entrata - Multifamily & Affordable Housing SaaS

Oct 2020 - Jul 2024

Senior Software Engineer · Business Analysis & Delivery · 100K+ MAU

- **Gathered and documented business requirements** for 20+ features across affordable housing and property management modules - translated stakeholder needs into functional specs delivered across 120+ Agile sprints.
- Analyzed workflows and data to identify gaps: built Python/Bash automation that eliminated 80% of manual operational work after mapping current-state processes and defining future-state requirements.
- Created **compliance dashboards** (Power BI, SQL) for LIHTC/HUD tax credit unit compliance - property managers used these to track certification status and eligibility across portfolios.
- Optimized 200+ SQL queries (30% performance gain); authored stored procedures for eligibility, rent calculation, and regulatory reporting used by operations and compliance teams.
- Coordinated with leasing, compliance, and technical teams across full project lifecycle - planning, testing, UAT, and production release; documented configurations and rollback procedures for each cycle.
- Presented data-driven findings and process improvement recommendations to stakeholders; mentored 5+ engineers on analytical and documentation standards.

Entrata India

Sep 2018 - Oct 2020

Junior Software Engineer · Pune, India

- Analyzed and documented requirements for affordable housing module workflows; delivered data reconciliation pipelines for 1.6M HUD/Tax Credit records with structured validation logic.
- Created process documentation and user guides for 56 state-specific housing certification workflows used by operations teams nationwide.

PROJECTS

AgentProbe - Chatbot & LLM Evaluation Toolkit

Chatbot Evaluation · Structured Assessment · AI

Python · FastAPI · LangGraph · GitHub Actions · Anthropic · OpenAI · Azure OpenAI · Groq · Ollama - github.com/venkatalakshmiparimala/AgentProbe

Built a structured chatbot and LLM evaluation framework that compares outputs across five AI providers against defined assessment criteria. Scores each response on five dimensions: factual accuracy, hallucination rate, retrieval quality, completeness, and compliance - the same evaluation framework this role requires. Evaluation results tracked via CI regression tests with golden datasets; automated reporting flags quality drops before release. Used to evaluate 1,240+ chatbot responses across test cases.

CrashClarity - Data Analysis & Reporting (104K Records)

SQL · Dashboards · Reporting

Python · R · SQL · scikit-learn · Tableau · GitHub Pages

End-to-end business analysis pipeline: gathered data requirements, mapped analysis workflow, built SQL-driven data preparation, applied Random Forest and Logistic Regression models (85%+ accuracy), and published an interactive dashboard communicating insights to non-technical stakeholders.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia May 2026

GPA 3.87 · Business analytics, data pipelines, statistical modeling, process optimization, AI systems

B.Tech Computer Science & Engineering - JNTU Anantapur 2017

TECHNICAL SKILLS

Analysis & Reporting: SQL (PostgreSQL, Oracle, MySQL), Python (Pandas, NumPy), R, Excel (advanced), Power BI, Tableau, ggplot2, Quarto, process mapping, gap analysis

AI & Evaluation: LLM evaluation frameworks, chatbot output scoring, LLM-as-judge, golden dataset CI, prompt engineering, Anthropic Claude, OpenAI, LangGraph, RAG pipelines

Methodology: Agile/Scrum (120+ sprints), requirements gathering, functional specifications, UAT, KPI tracking, risk management, documentation

Tools & Platforms: Jira, GitHub, Salesforce, Microsoft Office Suite (Word, Excel, PowerPoint, Teams, SharePoint), AWS, Docker